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Mars Global Mosaic: Mapping Lava Flows, Fluvial Channels, and Impact Craters in Visible and Infrared

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ABSTRACT

Viking MDIM 2.1 visible colour, THEMIS daytime and nighttime thermal-infrared mosaics and the HRSC/MOLA blended elevation model were co-registered on a single 100 m and 200 m grid that spans ±60° latitude, 86.6% of the surface of Mars, and the stack was used to map impact craters, fluvial channels and lava flows. Over the Ius Chasma type area the three visible bands are one dimension ($r = 0.93$–0.99) while day and night infrared are nearly independent of them and of each other ($|r| \leq 0.12$ for night). Across ±60°, however, the held THEMIS mosaics are 8-bit products contrast-stretched region by region: dusty and rocky provinces differ by 47.67 DN in Viking red and by −0.19 and +0.02 DN in day and night infrared, and against calibrated TES thermal inertia the THEMIS pair keeps only a weak local signal ($|r| \approx 0.1$) while Viking albedo tracks inertia planet-wide ($r = -0.36$). Fill depth on the elevation model yields 465,354 closed depressions ≥ 1 km; against the Robbins crater database 62.2% of those ≥ 5 km are catalogued craters, but only 15.3% at ≥ 1 km, and recall levels off near one half because breached craters are invisible to the method. A routed network of 969,822 channel segments has 89.8% of its length on slopes under 2°, and its stream order is set by the accumulation threshold rather than by the terrain. A support vector machine trained on 512 hand-drawn landform polygons and scored on whole 15° blocks it never saw reaches 73.8%, κ 0.58 against 58.6% for a one-class map; the thermal bands add a resolved 1.4 to 2.0 points. A U-Net trained on the same labels reaches 98.9%, κ 0.98 as a map of terrain provinces. No band separates the lava contacts of the 1:20 M geologic map at Athabasca Valles much better than chance, so lava flows are taken from that map. Three of four hypotheses are not supported, and the fourth was answered by another method.

Keywords: Mars — Planetary surfaces — Remote sensing — Infrared imaging — Impact craters — Lava flows — Fluvial geomorphology — Machine learning

1. INTRODUCTION

Athabasca Valles, a young channel system south of Elysium, was read for decades as an outflow channel carved by water released from the Cerberus Fossae (D. M. Burr et al. 2002; J. B. Plescia 2003). High-resolution imaging later showed the channel floor draped by a single turbulent flood of lava (W. L. Jaeger et al. 2007, 2010), and the USGS geologic map places it in the Late Amazonian volcanic unit *lAv* (K. L. Tanaka et al. 2014). The case makes the question of this project concrete: can a co-registered visible and thermal-infrared mosaic separate lava, water and impact landforms, and how much of the answer comes from the thermal infrared?

Thermal infrared is the natural candidate. Rock and coarse debris hold heat through the night while dust cools quickly, so the diurnal temperature contrast follows thermal inertia, a property of particle size and induration (M. T. Mellon et al. 2000; R. L. Fergason et al. 2006; N. E. Putzig & M. T. Mellon 2007). Visible albedo also separates bright dust from dark basalt (S. W. Ruff & P. R. Christensen 2002), and topography carries the morphology of craters, channels and flow fronts. The global mosaics that hold each of these quantities have been available for years (B. A. Archinal et al. 2003; C. S. Edwards et al. 2011; R. L. Fergason et al. 2018), but they come on three different coordinate frames, and a usable stack of them is not a given.

This paper reports the stack, the landform products made from it and the tests run on them. The goals were (1) one co-registered stack of Viking visible colour, THEMIS day and night infrared and HRSC/MOLA topography on a single grid; (2) maps of the three landform families, first in a type area and then across the

whole extent; (3) a test of whether the thermal infrared separates surfaces that visible light cannot; and (4) a score for every product against something it never saw. Four hypotheses were set when the project was proposed: H1, that day infrared shows lava-flow margins under dust that Viking misses; H2, that fluvial channels have shallower gradients than volcanic ones and the two separate on gradient against thermal response; H3, that the multi-band composite gives more stable unsupervised classes than any single input; and H4, that crater rims found from image gradients recover most of the 141 named craters larger than 100 km. Each test below wrote its prediction and verdict rule into its script before it ran.

2. DATA

2.1. *The source mosaics*

Four global or near-global products were used (Table 1). The THEMIS mosaics (P. R. Christensen et al. 2004; C. S. Edwards et al. 2011) are single-band 8-bit display products: the night mosaic's label gives a base of 0 and a multiplier of 1, with no scaling to radiance or temperature, and the two mosaics were stretched independently (2nd and 98th percentiles of 35 and 205 DN by day, 58 and 203 DN by night over Ius Chasma). Calibrated thermal inertia cannot be derived from them. Viking MDIM 2.1 has every one of its 4.25 billion pixels valid; the HRSC/MOLA blend (D. E. Smith et al. 2001; R. Jaumann et al. 2007; R. L. Fergason et al. 2018) is signed 16-bit metres on a geographic grid.

2.2. *Reference data*

Four independent references score the products (R. G. Congalton 1991). The USGS Geologic Map of Mars, SIM 3292 (K. L. Tanaka et al. 2014), supplies 1,311 unit polygons and is the lava-flow reference. The Robbins and Hynek database (S. J. Robbins & B. M. Hynek 2012), 2020 release, holds 385,049 craters ≥ 1 km. TES thermal inertia (N. E. Putzig & M. T. Mellon 2007), dayside and nightside at 20 pixels per degree, is the only calibrated thermal product. The IAU gazetteer (IAU Working Group for Planetary System Nomenclature 2019) names 141 craters larger than 100 km. Every IAU crater larger than 100 km matches a Robbins crater with a median centre offset of 1.0% of its diameter, and the Athabasca Valles floor falls in unit *lAv* of the geologic map, as published.

2.3. *The analysis extent and the type areas*

No global THEMIS night mosaic exists: coverage stops at ±60°. The analysis stack was therefore bounded at ±60°, which removes a third of the rows of a simple cylindrical grid but only 13.4% of the surface: 86.6% of Mars lies within ±60°. Relief inside the band equals global relief (29,343 m, Hellas Planitia to Olympus Mons); the polar layered deposits and ice caps, none of them a target landform, are lost. Two type areas prove methods before they run at ±60°: Ius Chasma (271–286°E, 6–13°S), a canyon with the Louros Valles tributaries, and Athabasca Valles (150–162°E, 4–14°N), the flood lava (Figures 1 and 2).

3. METHODS

All processing ran headless through `arcpy` in ArcGIS Pro 3.7 (Esri 2026), with GDAL (GDAL/OGR contributors 2024) and NumPy (C. R. Harris et al. 2020) for raster arithmetic, on a workstation with an Intel i7-8700K, 64 GB of memory and an NVIDIA RTX 2080 Ti; figures were drawn with Matplotlib (J. D. Hunter 2007) or exported from the project's map layouts. Every product and figure has a script that regenerates it (L. M. Edwards 2026).

3.1. *Co-registration on one grid*

The four sources span three frames: THEMIS on a central meridian of 180° (longitudes 0–360°E), Viking on 0° (−180 to 180°E), and the DEM in geographic degrees. A THEMIS window read at face value lands half a planet away, and elevation derivatives computed on the degree grid carry an undefined vertical factor. A global Composite Bands run on the unharmonised sources was cancelled four times by 13 September (and a fifth time on 24 September); projected first and bounded to the type area, the same composite took 8.8 s.

The common grid is read from the night mosaic itself: equidistant cylindrical, central meridian 180°, sphere radius 3,396,190 m (B. A. Archinal et al. 2018), at 100 m (G100, 213,388 × 71,130, 15.18 billion pixels) and as its exact 2 × 2 aggregation at 200 m (G200, 106,694 × 35,565). The day mosaic sits on the night grid at an offset of exactly +1 and +17,783 pixels, so neither THEMIS mosaic is ever resampled. The DEM cell is 199.99999 m on this sphere, so a nearest-neighbour warp to G200 reproduces 262,144 of 262,144 sampled source pixels exactly, where a cubic warp alters 36% of them by up to 28 m; terrain is therefore computed on G200 and imagery on G100. Slope, aspect and hillshade are computed in metres on the projected grid, which removes the vertical-factor warning of the degree-grid products.

The DEM's last column proved defective (nodata in 24,335 rows and wrong values elsewhere; the file also ends 0.46 pixel short of +180°), which painted a 2 km stripe of spurious classes along 180°. It was replaced by

Table 1. The source mosaics, as held

Product	Cell	Size (px)	Frame	Holds
Viking MDIM 2.1 colour	231.5 m	92,160 × 46,080 × 3	cylindrical, CM 0°	visible albedo, RGB
THEMIS Day IR v12	100 m	213,390 × 106,696	cylindrical, CM 180°	day brightness, 8-bit
THEMIS Night IR v14	100 m	213,388 × 71,130	cylindrical, CM 180°, ±60°	night brightness, 8-bit
HRSC/MOLA blend DEM v2	0.0033741°	106,694 × 53,347	geographic, degrees	elevation, −8.5 to +21.2 km

Note: THEMIS 6.8–14.9 μm. All four are uncompressed and stored one scanline per block; together about 62 GB. The DEM cell is 199.99999 m on the 3,396,190 m sphere.

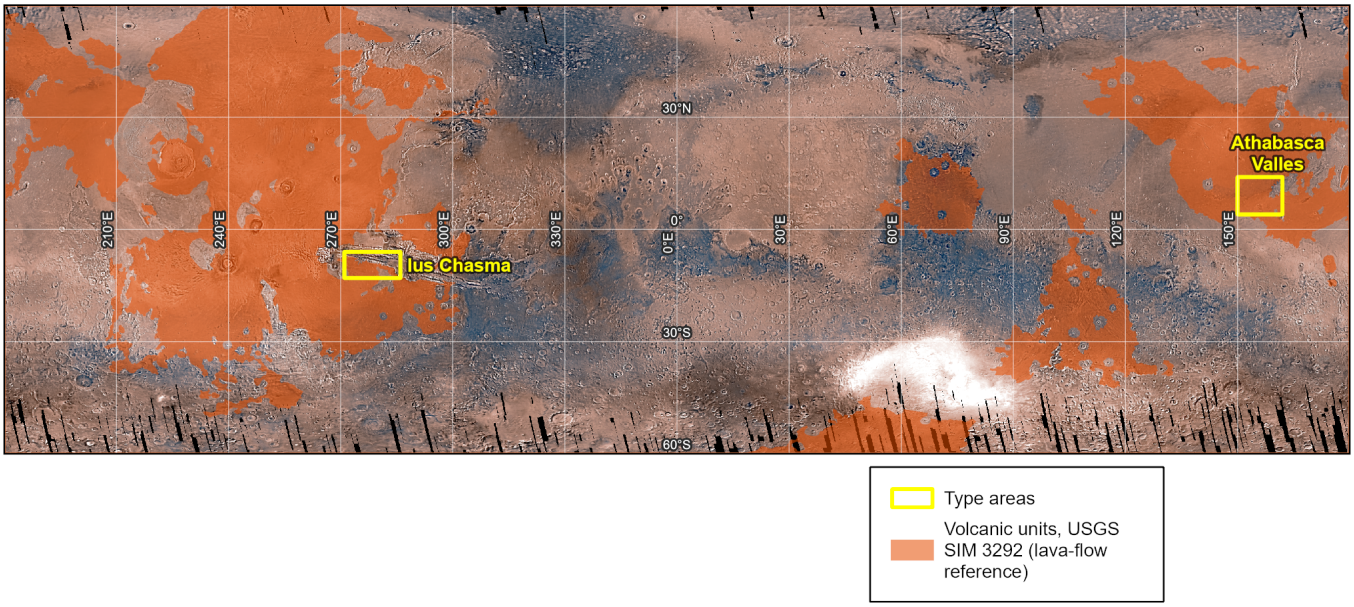


Figure 1. The analysis extent (±60°) over Viking MDIM 2.1 colour, the two type areas, and the volcanic units of the USGS geologic map SIM 3292 (K. L. Tanaka et al. 2014) in orange, the lava-flow reference of this study. Geographic coordinates, centred on 0°E.

the mean of its two neighbours and the affected products were recomputed over that column only.

3.2. *The classification stack*

The seven-band stack for classification is on G200 with 97.33% of pixels valid in every band: Viking R, G and B (bilinear from 231.5 m), THEMIS night and day infrared (2×2 averages on their own grid), slope in degrees, and local relief in a 9×9 window (1.8 km). Every band is stretched from its 1st and 99th percentiles over ±60° to 1–255. Elevation itself is excluded: a raw elevation band beside 8-bit bands dominates any distance-based classifier, and an earlier map built with one was 62% reproducible from elevation alone.

3.3. *The diurnal-contrast index*

Since calibrated inertia is unavailable, a relative index was formed: each THEMIS band is scaled by its own 2nd and 98th percentiles to $[0, 1]$ and night is subtracted from day, giving a dimensionless value in $[-1, 1]$ that is high for a large diurnal swing (dust-like) and low for a damped one (rock and coarse debris). It is valid only within the area over which it was stretched. Over Ius Chasma its mean is +0.0757 (s.d. 0.279), and it is nearly independent of topography ($r = +0.108$ with elevation, −0.248 with slope).

3.4. *Closed depressions and channels*

The `Fill` operation (S. K. Jenson & J. O. Domingue 1988) floods every closed depression, which is what a fresh crater is topographically, so fill depth is a crater detector. A depression is kept if it is deeper than 20 m, at least 1 km across (about ten 100 m pixels, the resolving rule used throughout), no more than twice as long as wide, and fills at least 0.55 of its bounding box. The thresholds were tuned on two named craters near Ius Chasma: Perrotin (82.8 km) is found at 77.0 km (−7.1%) and ranks first of 1,321 candidates; Oudemans (124.2 km) is missed, because it is breached into the canyon and its lowest rim point lies 33 m below its floor.

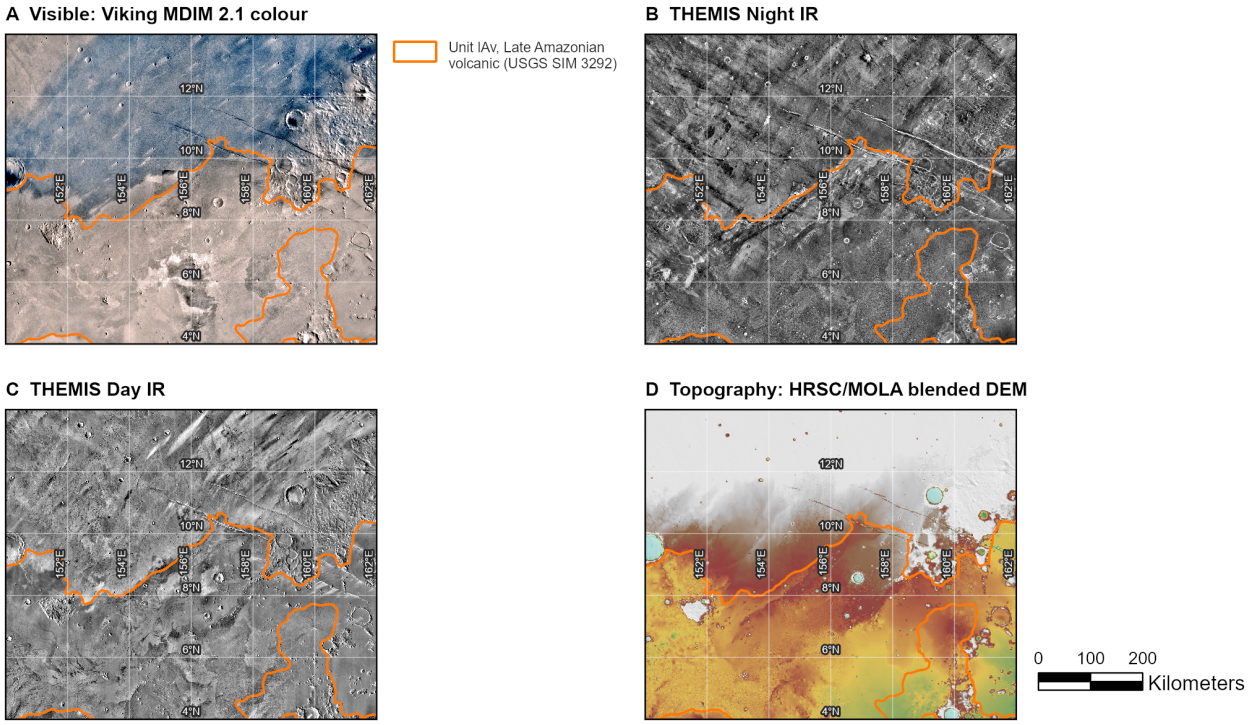


Figure 2. The Athabasca Valles type area on the 100 m stack: (A) Viking colour, (B) THEMIS night infrared, (C) THEMIS day infrared, (D) the HRSC/MOLA elevation model (−4,495 to 1,945 m) over hillshade. The orange outline is unit *lAv*, Late Amazonian volcanic, of SIM 3292.

Across ±60° the detector runs at two scales: the whole extent at 1.6 km for basins ≥ 20 km, and 8,192-pixel tiles of G200 with a 1,024-pixel (205 km) halo for depressions ≥ 1 km.

Channel centrelines come from D8 flow routing on the filled surface (S. K. Jenson & J. O. Domingue 1988) with a 5,000-cell (50 km^2) accumulation threshold, ordered after A. N. Strahler (1957). Because `Fill` turns closed basins into synthetic lakes, only stream cells where the fill raised the surface by at most 1 m are kept.

3.5. *Supervised classification and its score*

Five hundred and twelve polygons were drawn by hand in four landform classes across the planet: 302 Crater, 44 steep or windy hills, 44 lava tube and 122 Normal Ground. Clipped to ±60°, they were split by whole 15° × 15° blocks (14 of 39 blocks held out) into 329 training and 183 held-out polygons, each polygon whole and assigned to the block holding its centre (D. R. Roberts et al. 2017); one held-out polygon overlaps a training one by 51.5 km^2 across a block edge. A support vector machine (C. Cortes & V. Vapnik 1995) was trained on the training polygons (1,000 samples per class) and applied to all of ±60° at 200 m in 21 tiles of 16,384 pixels.

Accuracy is the area-weighted (by cos latitude) share of held-out polygon area given its own class, with Cohen's κ (J. Cohen 1960) and per-class producer's and user's accuracy (R. G. Congalton 1991). Three baselines accompany every score: one class everywhere (58.6% on the held-out set), the share of the map that elevation alone reproduces (each 500 m bin given its majority class), and the share of named craters ≥ 5 km labelled Crater inside $0.7R$ against a ring at 1.5–$2.5R$. A majority filter removes speckle; its window was set from the resolving rule (about 2 km) rather than from the score, because every score rises with every window when the labelled polygons are large.

3.6. *Deep learning*

Two models were trained on the training blocks only and scored on the same held-out polygons. A RetinaNet detector (T.-Y. Lin et al. 2017) with a ResNet-50 backbone (K. He et al. 2016) was trained on 3,622 chips of 256 × 256 × 7 pixels (stride 128) holding 5,293 boxes; its score is per polygon, recall being the share of held-out polygons at least half covered by boxes of their own class. A U-Net (O. Ronneberger et al. 2015) with a ResNet-34 backbone, class balancing and unlabelled ground ignored in the loss, was trained for 30 epochs on the same chips as pixel labels, applied to all of ±60° in 243 tiles of 4,096 pixels with a 256-pixel margin, and scored with the same code as the support vector machine. Because the classification tool works internally in

Table 2. Band correlation over Ius Chasma (8,891 × 4,150 pixels at 100 m)

	R	G	B	Day	Night
Viking R	1	0.960	0.925	−0.486	−0.121
Viking G		1	0.993	−0.496	−0.085
Viking B			1	−0.488	−0.068
Day IR				1	0.105
Night IR					1

256-pixel tiles, the U-Net was run a second time on a grid offset by 128 pixels, and each pixel was taken from the pass that places it farther from a tile edge. Both models used the Deep Learning Libraries of ArcGIS Pro, built on PyTorch (A. Paszke et al. 2019).

3.7. *Uncertainty*

Intervals are 95% paired bootstrap intervals (B. Efron & R. J. Tibshirani 1993) over whole held-out polygons (1,000 resamples), over contacts for the flow-margin test and over 50 km cells for the channel test, so that spatially correlated pixels are never treated as independent.

4. RESULTS

4.1. *The mosaic and the case for fusion*

Figure 3 shows the four layers on the common grid. Over Ius Chasma all five bands are valid on 100% of pixels, and their correlations (Table 2) give the measured case for the fusion: the three Viking bands are one dimension, day infrared is anticorrelated with visible ($r \approx -0.49$; bright dust is cool by day), and night infrared is the most independent band. The effective dimensionality is about three. At Athabasca Valles the pattern repeats (night $|r| \leq 0.15$ against every band). At ±60° the registration was confirmed between Viking red and day infrared: the strongest correlation within ±19 km lies at zero shift ($r = -0.147$), and none survives a 180° roll.

4.2. *What the thermal pair can and cannot do*

Inside one area the diurnal-contrast index behaves as a material signal should. On the object-based classes of Ius Chasma the steep canyon wall (mean slope 21.5°) and the mantled chasma floor (17.0°) are the two steepest units, yet differ by 0.46 in the index at opposite signs, and fresh crater ejecta come out damped against the plateau.

Across ±60° the signal does not survive (Figure 4). Over seven provinces whose thermal-inertia contrast is well known, the dusty group (Arabia Terra, Amazonis, Tharsis) and the rocky group (Syrtis Major, Acidalia, Hellas floor, Valles Marineris) differ in the index by −0.0009, against a scatter of 0.13 to 0.23 inside each province. The sources show why: every province reproduces the mosaic-wide THEMIS mean and spread (125.75±34.36 DN by day), so dusty and rocky provinces differ by −0.19 DN by day and +0.02 DN by night, while Viking red, which is not stretched that way, separates them by 47.67 DN. TES measures a fivefold contrast between the same groups (dayside medians of 45 and 246 tiu). The THEMIS DN are local contrast, not radiance.

Calibrated inertia agrees (Figure 5). Averaged into TES cells (∼ 3 km), Viking red tracks log thermal inertia across ±60° ($r = -0.31$ dayside, −0.36 nightside), much less within a 15° block (median −0.06, −0.10): a regional signal, the dust provinces. Night infrared and the index carry a weak local signal in the physical direction (median within-block $r = +0.11$ and −0.08 against nightside inertia, quartiles of one sign) that vanishes planet-wide, as region-by-region stretching predicts; day infrared carries none. In the type areas the index tracks nightside inertia weakly at Ius ($r = -0.22$) and not at all at Athabasca (−0.01), where Viking albedo does (−0.64). The prediction that the index would track inertia clearly in both windows fails.

4.3. *Craters*

The basin pass finds 217,622 closed depressions at 1.6 km in 48 s, 5,144 of them ≥ 20 km, including Hellas (2,697 km across, 7,857 m deep), Utopia (2,093 km) and Isidis (1,026 km). Of the 117 named IAU craters ≥ 100 km inside ±58°, 88 (75%) have a closed depression near their centre and 79 (68%) one whose diameter is within 50%; on those the median diameter error is +2.2% (10th and 90th percentiles −12.6% and +20.8%). This answers H4 with fill depth rather than image gradients.

The fine pass at 200 m, 11 h on the workstation, yields 465,354 closed depressions ≥ 1 km with clean tile seams (density across seams 0.990 and 0.979 of mid-tile). Against Robbins (Figure 6), a candidate matches if its centre lies within half the catalogued diameter and its diameter within a factor of two; a chance baseline shifts every candidate east by three of its own diameters. Precision rises from 15.3% at ≥ 1 km to 62.2% at ≥ 5 km and 83.1% at ≥ 20 km, against chance below 11% at every size; recall levels off near one half, because breached and degraded craters are not closed. In the type areas the same rule gives a precision of 11% (Ius) and 13% (Athabasca) at ≥ 1 km.

Mapped by density, the small candidates do not follow the craters: per 0.5° cell their log density correlates with that of Robbins at $r = -0.11$, nearly flat across latitude

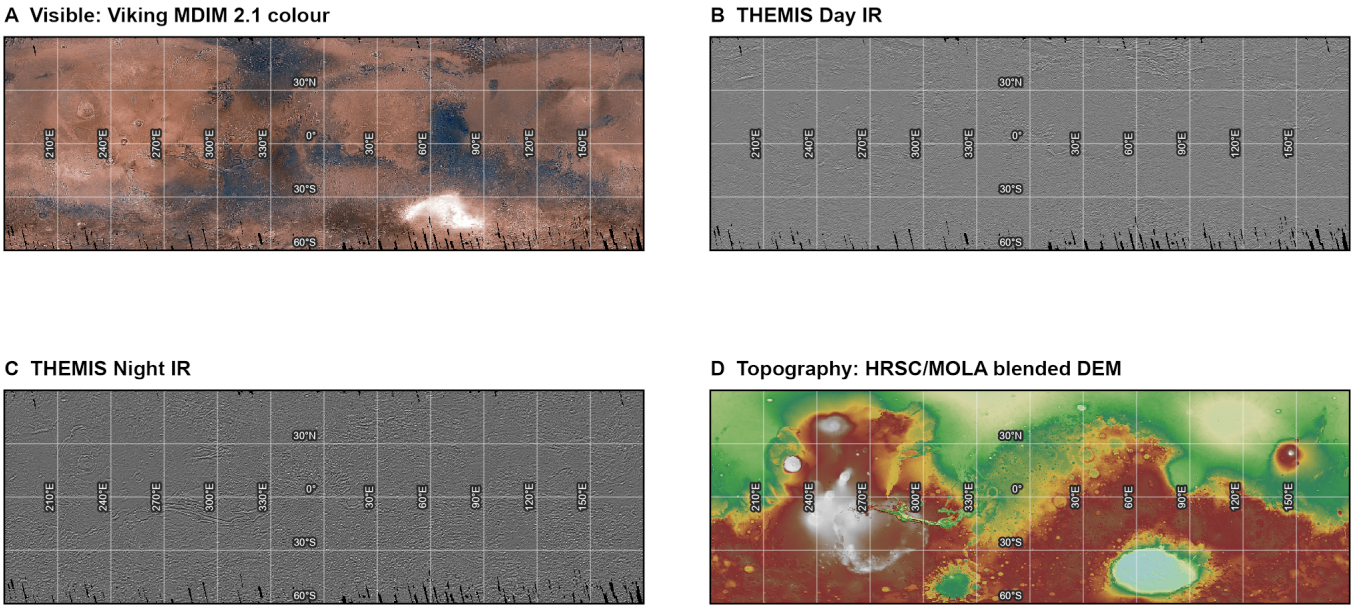


Figure 3. The co-registered mosaic at ±60°, drawn from the 200 m stack: (A) Viking visible colour, (B) THEMIS day infrared, (C) THEMIS night infrared, (D) HRSC/MOLA topography over hillshade. Both THEMIS panels look nearly uniform at this scale because the mosaics are contrast-stretched region by region (Section 4.2).

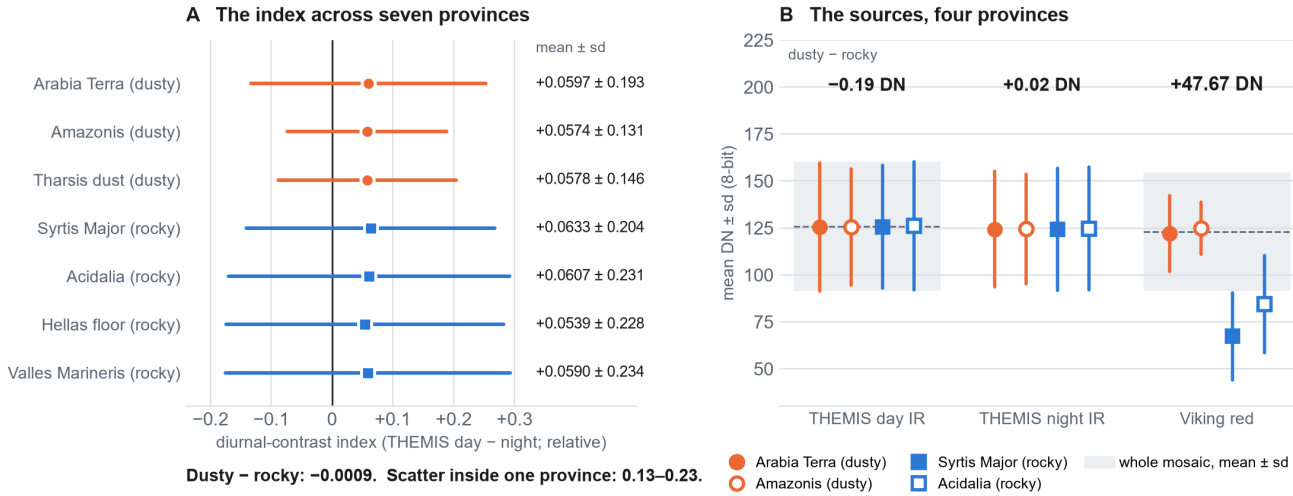


Figure 4. (A) The diurnal-contrast index over seven provinces: dusty and rocky provinces differ by −0.0009. (B) The source DN of four provinces: the THEMIS mosaics repeat the mosaic-wide mean and spread everywhere, while Viking red keeps a 47.67 DN albedo difference.

bands while Robbins doubles, and their pattern shows north–south stripes where the HRSC/MOLA blend is sharpest. Below about 5 km the detector maps the elevation model's resolution as much as the craters. Relative crater density per geologic unit, which needs no detector, ranks the units as expected: on *lAv* Robbins counts 1,122 craters ≥ 1 km per 10^6 km^2, against 1,760 and 2,183 on the older units of the Athabasca window; the detector reverses the order (4,966 on *lAv*) by counting non-crater pits on flat lava. These densities are relative and are not converted to ages (W. K. Hartmann 2005).

4.4. *Channels*

At Ius Chasma, `Fill` raised 21.7% of the scene by more than 1 m and the canyon floor by up to 2,077 m, and 56.1% of the first routed network ran across that synthetic surface (Figure 7). It looked plausible: long, smooth, dendritic trunks down the chasma. With the fill mask, 2,610 candidates (15,428 km) remain; 64% lie

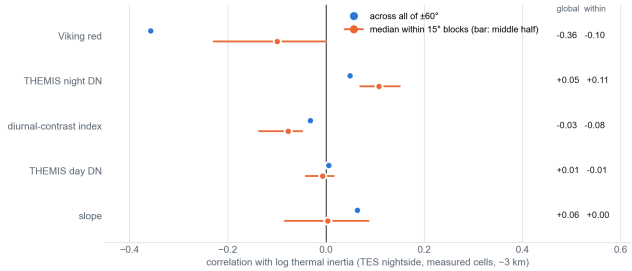


Figure 5. Correlation of each ±60° layer with log TES nightside thermal inertia (N. E. Putzig & M. T. Mellon 2007) in measured cells (∼ 3 km): across the whole extent (blue) and the median within 15° blocks with its interquartile range (orange).

on plateau under 2° where routing follows noise, and 188 are steep (≥ 5°) and rock-floored (index < −0.15), the subset worth review.

At ±60° the fill mask discards 60.5% of stream cells and leaves 969,822 segments (6.17 million km; orders 1–5: 564,882, 360,236, 41,442, 3,165 and 97; Figure 8). Of 40,410 sampled midpoints, 89.8% lie on ground under 2°, and 96.7% of those of order 3 and higher: the long high-order paths gather on smooth ground where the DEM holds few pits, and their length per cell correlates negatively with the density of 1–2 km depressions ($r = -0.35$). The project holds no channel reference at this extent, so the network is not scored; a valley-network map such as that of B. M. Hynek et al. (2010) would be the natural reference for a later test.

How many tributary orders the 100 m stack resolves (test T7) has no answer the elevation model supports. Over thresholds from 1 to 500 km^2 at Ius Chasma, the highest order falls from 5 to 2, about one order per tenfold rise in the threshold, while only 30.8–36.7% of first-order cells lie on slopes ≥ 2° (53.3% of the window does); the rule set in advance (first-order streams at least half on ≥ 2°) is met by no threshold.

H2 was tested at a first stage with the window standing in for origin (Ius fluvial, Athabasca volcanic). The volcanic window's 3,272 candidates are shallower, not steeper, than Ius's 2,588 (median 1.4 against 10.2 m km^{-1}, difference −8.85 m km^{-1}, interval −15.69 to −5.76), and the thermal z-score separates the two no better than chance (AUC 0.50, against 0.81 for gradient). What separates them is the setting, chasma walls against lava plains, so stage one cannot settle H2; stage two waits for reviewed channels with their origin set.

4.5. *Lava flows*

The classification's lava-tube class does not find lava plains. Against SIM 3292 unit groups, expressed as enrichment $P(\text{class} \mid \text{group})/P(\text{class})$ (Figure 9), lava tube is slightly depleted on volcanic plains (×0.84) and enriched on volcanic edifices (×3.30) and aprons (×4.29): steep volcano flanks. Normal Ground covers 76% of the volcanic plains. The other classes sit where the geologic map says they should, so the map is coherent as a terrain map but is not a lava-flow map. A relabelled class trained on the map's volcanic units fared worse: as a fifth class it lowered the other three classes' held-out accuracy from 70.2% to 64.6%, and only 30% of what it called volcanic was, below the 38% base rate. Lava flows in this study therefore come from the geologic map's volcanic units.

H1 was tested on the 14 internal contacts (2,764 km) between the five SIM 3292 units of the Athabasca window. Profiles every 2 km, ±20 km perpendicular to each contact, count as separating when the contrast across them exceeds the 90th percentile of 2,000 random profiles, so chance is 10% (Figure 10). Across the 905 profiles on the ten contacts of *lAv*, day infrared separates 21.0% (interval 3.3–38.4%), Viking red 12.3%, night infrared 11.7% and the index 13.0%. Day infrared leans toward H1 by +8.7 points over Viking red, but the interval (−1.7 to +18.1) includes zero, so by the rule set in advance H1 is not supported. A map at 1:20 M generalises contacts by kilometres, and ten contacts cannot settle the question; H1 needs margins digitised on the 100 m stack.

4.6. *The landform classification*

The published map (Figure 11) is the 200 m support vector machine map with an 11 × 11 (2.2 km) majority filter. On the 183 held-out polygons it scores 73.8%, κ 0.583, against 58.6% for one class everywhere and 69.9%, κ 0.532 unfiltered (Table 3, Figure 12). Normal Ground is found almost everywhere it was drawn (producer's 99.2%) and Crater is reliable where mapped (user's 93.3%), but lava tube is not a usable class (user's 2.4%). Halving the cell from 400 m bought no held-out accuracy (70.5% raw at 400 m). The map is no longer an elevation map (elevation bins reproduce 55.4%, within a point of its largest class alone, 54.7%), shows no tile seams (ratio 1.00), and labels 56.8% of the ground inside named craters as Crater against 46.2% in the ring around them.

Whether the thermal bands help (test T1) was measured twice, with the prediction of little or no gain written first. A Gaussian maximum-likelihood ablation on 2.69 million held-out pixels (Figure 13) finds that the thermal bands add +1.3 points (interval +0.6 to +2.2), that slope and relief alone (68.6%) match all seven bands (67.0%), and that without terrain the visible and thermal bands score below the one-class baseline (42.4%).

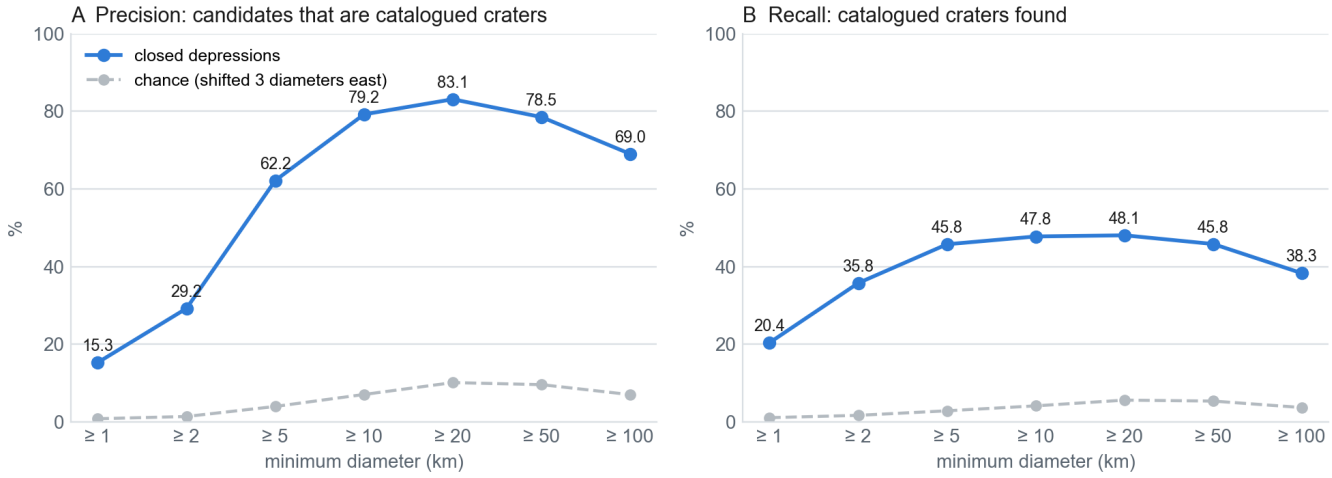


Figure 6. The 465,354 closed depressions ≥ 1 km against the Robbins and Hynek crater database (S. J. Robbins & B. M. Hynek 2012) by minimum diameter: (A) precision, (B) recall, each beside a chance baseline made by shifting every candidate three of its diameters east.

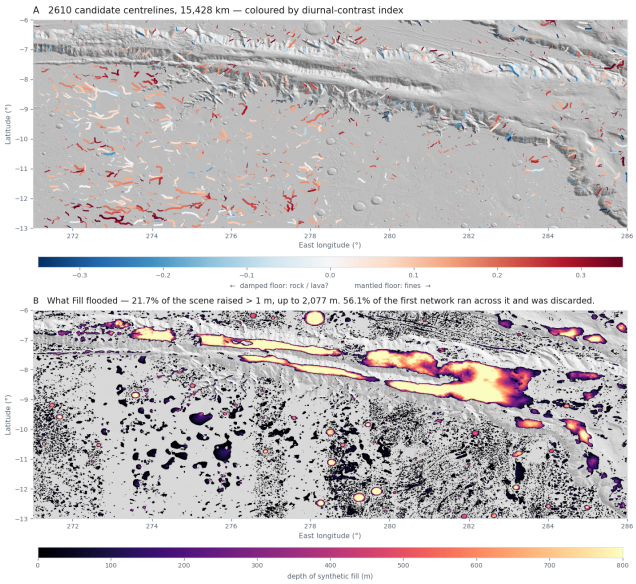


Figure 7. Channel candidates at Ius Chasma. (A) The 2,610 centrelines kept after the fill mask, coloured by the diurnal-contrast index. (B) Where `Fill` flooded the surface: 21.7% of the scene raised by more than 1 m, up to 2,077 m; 56.1% of the first network ran across it and was discarded.

Retraining ArcGIS Pro's support vector machine over all of ±60° without each band group confirms it (Table 4): the thermal gain is the one difference resolved at both cells. It is not uniform: thermal helps Crater (producer's +1.9 points, nine more polygons won) and hills (+5.5 points) and hurts lava tube (−8.5 points). Terrain alone ties the score but makes a poor crater map (34 of 124 crater polygons won, against 103). The hand-drawn classes are landform classes, a terrain property, and cannot price a material signal.

Table 3. Held-out scores of the landform maps (183 polygons)

Map	Accuracy	κ
One class everywhere	58.6%	–
SVM, 400 m, raw	70.5%	0.540
SVM, 200 m, raw	69.9%	0.532
SVM, 200 m, 5 × 5	72.0%	0.560
SVM, 200 m, 11 × 11 (published)	73.8%	0.583
U-Net, two passes	98.9%	0.980

Note: Area-weighted by cos latitude, scored at 400 m.

Table 4. Test T1 with ArcGIS Pro's support vector machine

Cell	Left out	Gain (pt)	95% interval
200 m	thermal	+1.36	+0.46, +2.47
400 m	thermal	+1.97	+1.08, +3.35
400 m	visible	−4.26	−10.96, +2.18
400 m	visible and thermal	−0.30	−6.54, +7.73

Note: Gain is the held-out accuracy of the all-band map minus that of the map without the group; paired bootstrap over 183 polygons, 1,000 resamples.

4.7. *Deep learning*

The RetinaNet detector found 41.9% of held-out Crater polygons with 87.6% of its in-polygon boxes correct (always naming Crater gives 71.1%) and 22.7% of Normal Ground polygons at 56.9% precision, but never

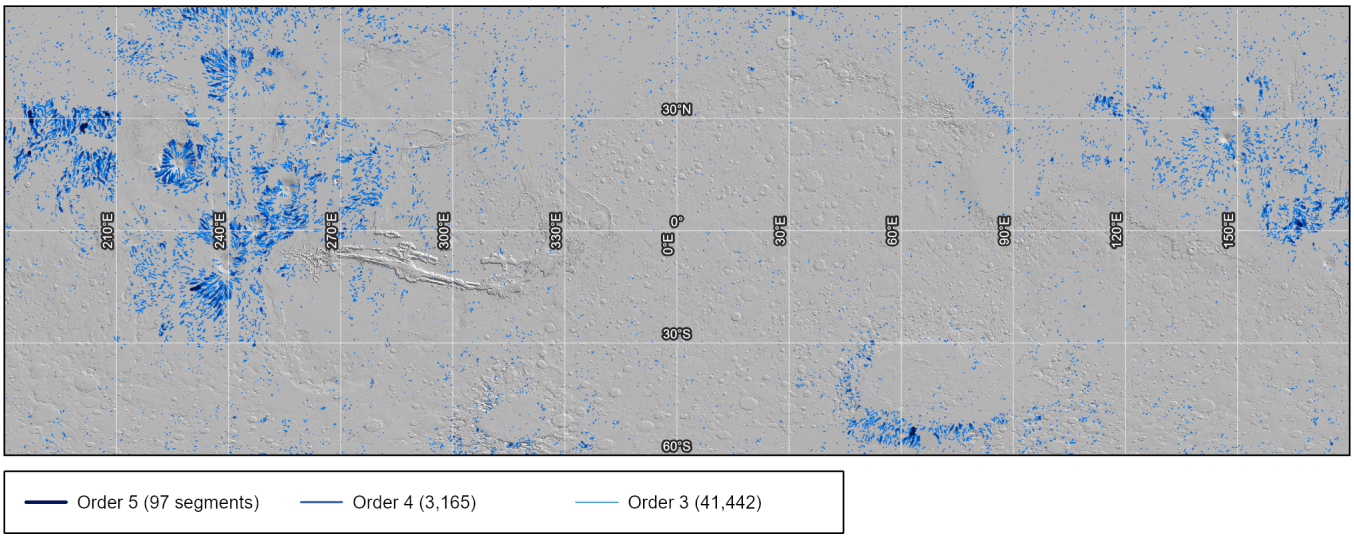


Figure 8. Routed channels of Strahler order 3 to 5 over the ±60° hillshade (orders 1 and 2 are not drawn). Most high-order paths lie on plains under 2° and follow the DEM's makeup as much as real drainage.

	area	Crater	steep/windy hills	lava tube	Normal Ground
Volcanic (v)	22.8%	×0.46	×0.34	×0.84	×1.38
Volcanic edifice (ve)	1.1%	×1.23	×1.19	×3.30	×0.46
Volcanic field (vf)	0.6%	×0.31	×0.34	×1.55	×1.36
Highland (h)	41.1%	×1.34	×0.95	×1.02	×0.79
Highland massif (hm)	2.8%	×1.65	×2.51	×1.79	×0.41
Highland undivided (hu)	2.1%	×1.18	×6.44	×1.22	×0.66
Impact (i)	5.5%	×1.39	×1.56	×1.17	×0.71
Apron (a)	1.3%	×1.02	×4.12	×4.29	×0.31
Transition (t)	7.9%	×1.05	×1.64	×0.71	×1.00
Transition undivided (tu)	2.4%	×0.88	×1.04	×1.28	×1.03
Lowland (l)	9.2%	×0.25	×0.05	×0.04	×1.66
Polar (p)	< 0.1%	×0.39	×0.00	×0.09	×1.57

Enrichment
P(class | group) ÷ P(class)
under ×0.5
×0.5–0.8
×0.8–1.25
×1.25–2
over ×2

Figure 9. The published landform map against the unit groups of SIM 3292: how much more often each class occurs in a group than across ±60° (×1: no association).

named hills or lava tube (Figure 14, left). Retrained with the backbone unfrozen for 30 epochs and with extra chips of the rare classes, it named hills (25.0% recall, 100% precision) but lost Normal Ground (4.5% recall) and over-predicted lava tube on craters; by the rule written before the run it did not improve. Boxes suit compact objects (C. Lee 2019) rather than area classes, and the whole-chip boxes of plains limit this model.

The U-Net is a different matter. In the held-out windows it scored 98.75%, κ 0.977, and over all of ±60° 98.8%, but with seams at its 256-pixel tiles (phase ratio 8.6–9.2 against about 1.0 for the support vector machine maps). The second, offset pass lowered the ratio to 1.2 and changed nothing else measured: 98.9%, κ 0.980 held out, with lava tube at 77.1% user's accuracy against 2.4% (Figure 14, right, and Figure 15). The map is a map of terrain provinces, not of features: the southern highlands are Crater, the northern lowlands Normal Ground, lava tube lies on Tharsis and Elysium and hills along Valles Marineris. It labels 91.3% of the ground inside named craters as Crater, but also 82.6% of the ring around them, and elevation bins reproduce 68.0% of it. The held-out labels mark areas that are mostly one class at 50 km, and a model that sees 51 km of context reproduces those areas; the score measures agreement with the hand-drawn areas at province scale. The support vector machine map therefore remains the published landform map, with the U-Net map beside it as a province map.

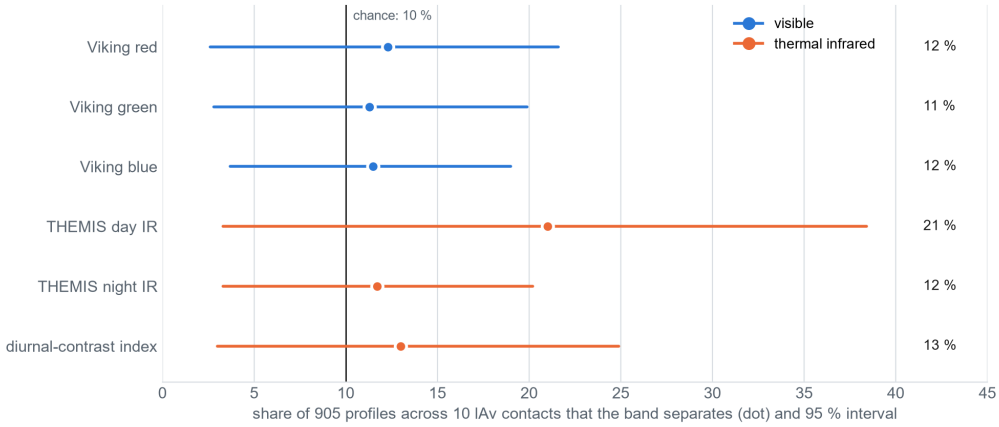


Figure 10. Test T5 (H1): the share of profiles across the geologic map's contacts at Athabasca Valles that each band separates, with bootstrap intervals over contacts; 10% is chance by construction.

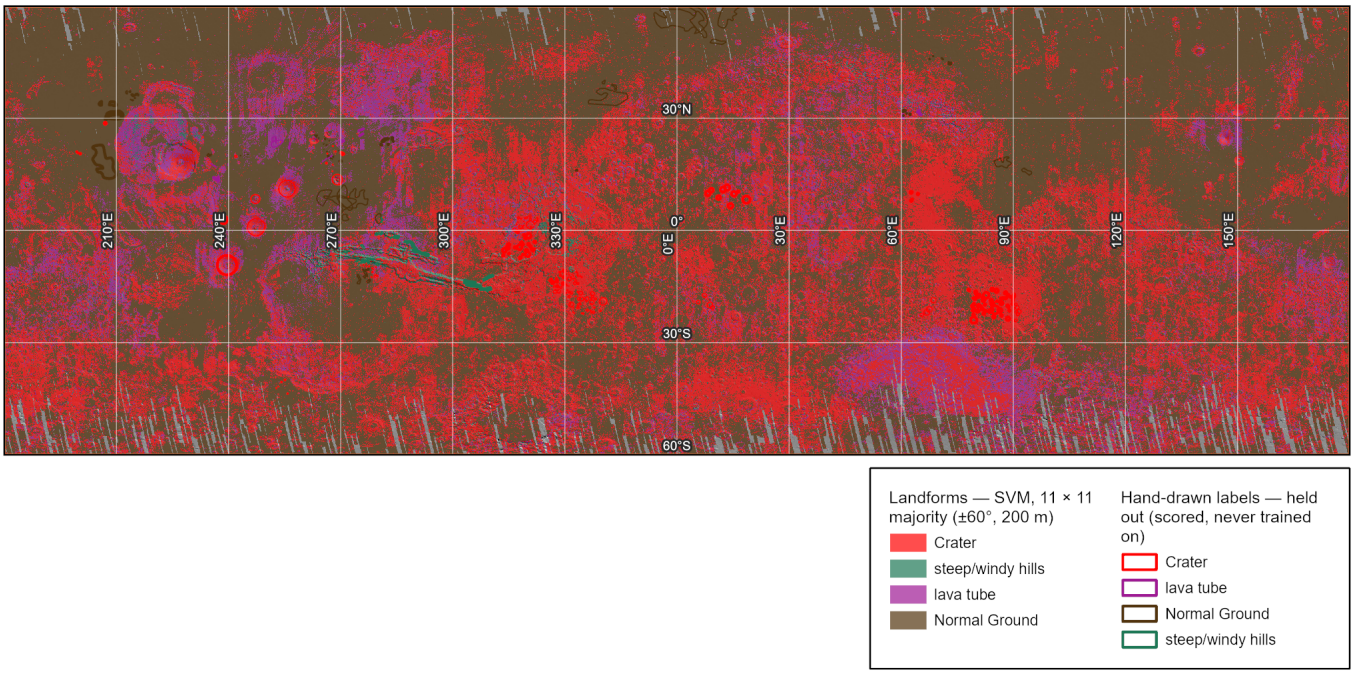


Figure 11. The published ±60° landform map: support vector machine on the seven-band 200 m stack, 11 × 11 majority filter, with the held-out hand-drawn labels outlined in their class colours. Held out: 73.8%, κ 0.58.

5. DISCUSSION

5.1. *The hypotheses*

Table 5 collects the verdicts. None of the four hypotheses is supported as stated. H3 failed outright: Iso Cluster partitions trained on the west and east halves of Ius Chasma agree less for the four-band composite (adjusted Rand index 0.40, L. Hubert & P. Arabie 1985) than for any single input (0.61 to 0.89), although the composite gives the most coherent maps; a one-band clustering is close to a density slice and reproduces easily. H1 and H2 were tested on stand-ins (map contacts and windows) and remain open for features digitised on the stack.

5.2. *Limits*

The central limit is in the data held. The THEMIS mosaics used here are display products, stretched region by region, so they carry local contrast and texture; planet-wide material discrimination needs calibrated products such as the TES inertia map, at the cost of resolution (3 km), or THEMIS-derived inertia (R. L. Fergason et al. 2006) where it exists. In this study the visible albedo carries the regional material signal.

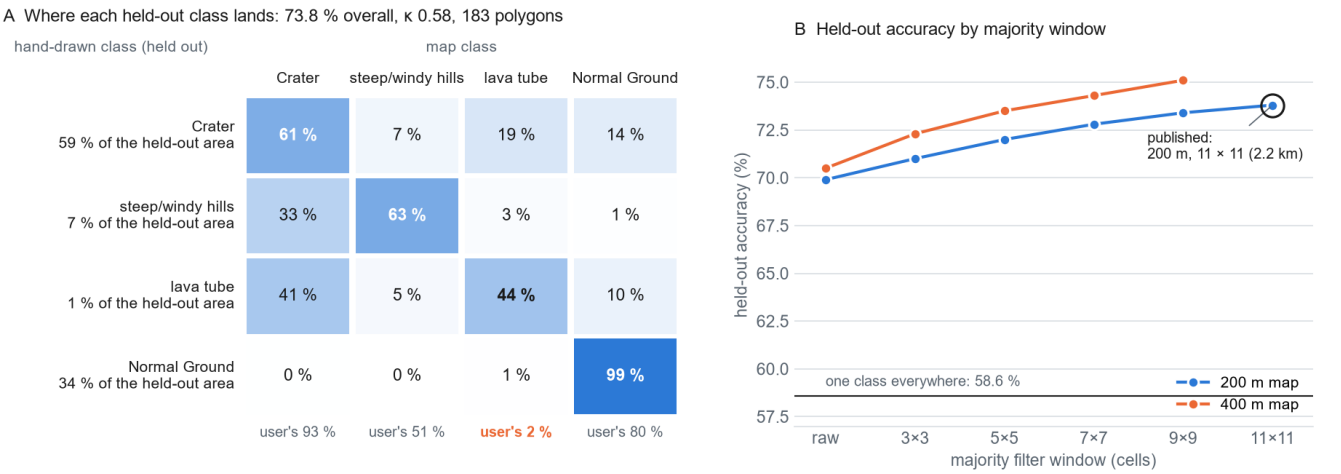


Figure 12. (A) Where each held-out class lands on the published map, as a share of its area. (B) Held-out accuracy against the majority-filter window for the 200 m and 400 m maps; the score rises with every window, so the window was set by feature size.

Table 5. The hypotheses and tests

	Claim	Verdict	Decisive number
H1	Day IR shows flow margins Viking misses	not supported at 1:20 M	day − Viking +8.7 pt, interval −1.7 to +18.1
H2	Fluvial and volcanic channels separate	not supported at stage 1	volcanic shallower; thermal AUC 0.50
H3	Composite classes are most stable	not supported	ARI 0.40 against 0.61–0.89
H4	Large IAU craters recovered	answered by fill depth	68% of 117 craters ≥ 100 km
T1	Thermal IR improves the classification	yes, slightly	+1.36 and +1.97 pt, intervals above 0
T3	THEMIS tracks calibrated inertia	weakly, locally	within-block $\lvert r\rvert \approx 0.1$; Viking −0.36
T7	Tributary orders resolved at 100 m	no DEM-supported answer	order follows threshold

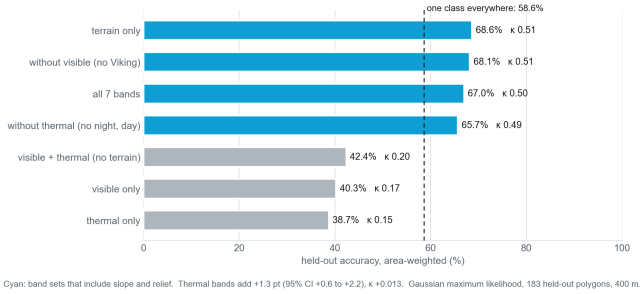


Figure 13. Test T1, first form: held-out accuracy of a Gaussian maximum-likelihood classifier on each band set at 400 m. The dashed line is one class everywhere.

The labels limit what the classification can show. The four hand-drawn classes are landforms, and slope and relief alone predict them as well as the full stack; a test of thermal value needs material labels, which only photo interpretation can give. Lava tube, with 44 polygons, is too thin a class for a planet-scale map; the U-Net makes it usable only as a province.

An audit of the 512 labels found no two classes sharing ground, but five Crater polygons drawn around volcanoes (Arsia, Pavonis and Ascraeus Mons, Tharsis and Ulysses Tholus) and six lava-tube polygons beside fluvial valleys. Arsia Mons alone is 31% of the held-out Crater area, so part of the Crater score, and of the U-Net's Crater province, measures volcanic shields. A seven-class scheme follows from the audit: Crater narrowed to impact craters, with volcanic edifice, apron and fluvial channel as new classes. Its labels await review; both maps will then be scored on the same held-out polygons, and the seven-class map replaces the published one only by a rule written before the run.

The detectors are elevation-model products. Closed depressions miss breached craters by construction, and below about 5 km their density follows the sharpness of the HRSC/MOLA blend. Channels inherit the same dependence, and the fill mask that removes synthetic lakes also breaks networks where the DEM is pitted.

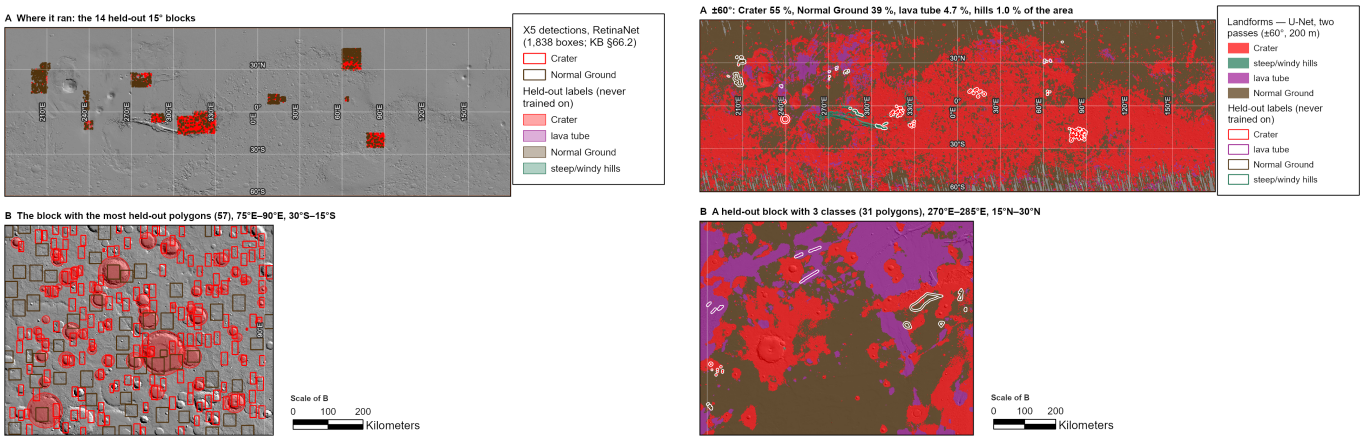


Figure 14. Left: RetinaNet detections in the 14 held-out windows, with the block holding the most held-out polygons (57, at 75–90°E, 30–15°S) enlarged. Right: the two-pass U-Net map of terrain provinces over ±60°, with the held-out block holding the most classes (270–285°E, 15–30°N) enlarged.

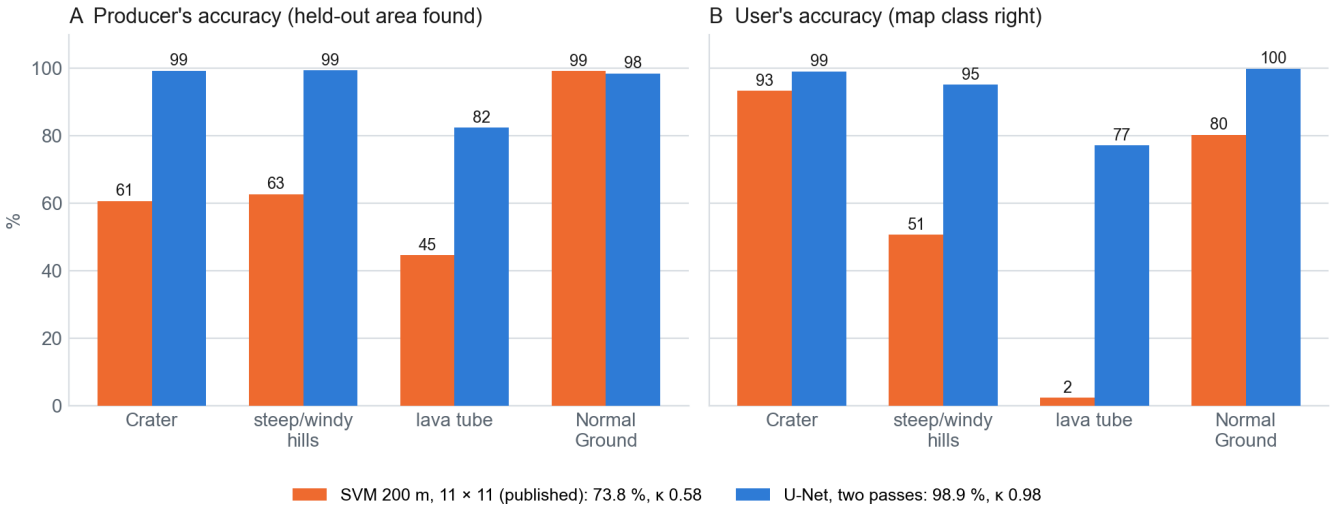


Figure 15. The published support vector machine map and the two-pass U-Net map under the same scorer: producer's (A) and user's (B) accuracy per class on the 183 held-out polygons.

The held-out test has been used once for each model. Any further tuning, of the deep-learning models above all, would turn the 14 held-out blocks into a validation set; a next round needs validation blocks of its own. Finally, the manual review of the 9,287 type-area candidates (2,610 and 3,283 channels, 1,685 and 1,709 depressions) has not started, so the digitised landform classes are empty and the machine products are presented as candidates with measured recall and precision.

6. CONCLUSIONS

1. Four global products now share one co-registered grid over ±60°, 86.6% of Mars, with the THEMIS pair never resampled and the elevation model reproduced exactly at 200 m.
2. Visible albedo, day infrared and night infrared are three nearly independent dimensions locally, but the held THEMIS mosaics carry local contrast only; calibrated thermal inertia follows Viking albedo across the planet.
3. Fill depth recovers 68% of the large named craters and gives 465,354 closed depressions ≥ 1 km whose precision against Robbins rises from 15.3% to 62.2% at 5 km; breached craters stay invisible.
4. Routed channels are dominated by the elevation model at ±60° (89.8% under 2°), and stream order is a parameter, not a measurement.
5. The published landform map scores 73.8%, κ 0.58 on blocks it never saw; the thermal bands add a

resolved 1.4 to 2.0 points. A U-Net reaches 98.9% as a terrain-province map.

6. Lava flows are not separable in these bands; the geologic map remains the lava-flow reference, and H1 and H2 await digitised margins and channels.

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Facilities: Mars Odyssey (THEMIS), Viking Orbiter, MGS (MOLA, TES), Mars Express (HRSC)

Software: ArcGIS Pro (Esri 2026), GDAL (GDAL/OGR contributors 2024), NumPy (C. R. Harris et al. 2020), Matplotlib (J. D. Hunter 2007), PyTorch (A. Paszke et al. 2019)

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